

Paper Information Record — Fang2022

Paper ID: 0018 | Oxford Missing Data Study

General Information

Field	Value
Paper ID	0018
Title	Anti-Corruption Campaign, R&D Efficiency, and Government Subsidies
Authors	Fang, Y., Kong, G., Nofsinger, J. R., & Wang, M.

Data Structure

Field	Value
Data type	Firm-year panel, two-way FE (firm + year), clustered SEs (firm)
Observations (N)	10,445 (regression sample after FE absorption and listwise deletion)
Preferred model column	Table 5, Panel A, Column 6 (firm+year FE, preferred spec)

Core Regression Model

Field	Value
Dependent variable	subsidy_s (government subsidy, standardized)
Focal IV	lrdefficiency_postremoval (R&D efficiency x post-removal DiD interaction)
Controls	lrdefficiency, laetc_s, lpostremoval, laetc_postremoval, lsoe, lpolitical, lroa, ltobinq, lleverage
Fixed effects	firm + year (two-way absorbed)
Clustered SEs	firm (CRV1)

Simulation Configuration

Field	Value
Key variables (3)	lroa, ltobinq, lleverage
MAR control	age (firm age, continuous, complete, correlated with key vars)
Mechanisms	MCAR, MAR, NMAR
Missingness proportions	1%, 5%, 10%, 20%, 30%, 40%, 50%
Methods	LD, Mean, Reg, Iter, RF, DL, MILGBM
Iterations per scenario	30
MAR/NMAR strength	1.5
M (multiple imputation)	5

Baseline Validation

Field	Value
Baseline coefficient	0.00595 (lrdefficiency_postremoval)
Baseline SE	0.00357

Field	Value
Baseline p-value	0.0959
Published value	0.006* (p<0.10, Table 5 Panel A Col 6) — exact match
R-squared	0.603

Run Log

Field	Value
Full run start	2026-04-13
Full run complete	2026-04-14
Total combinations	13230 (3x7x3x30x7)
Errors	0
DL method	TensorFlow 2.21.0, MLP 32->Dropout(0.1)->16->1, ReLU, Adam lr=0.005, EarlyStopping
MILGBM method	LightGBM 4.6.0, MICE M=5, Rubin's Rules, 30 estimators, max_depth=4
Note	Focal IV is borderline significant (p=0.096, one-star). B_prop~93% at MCAR 1pct for some combos is expected —